a particular transformer certain maximal number of consumer connection can only be linked. Now the question of deciding, whether a new Transformer/Substation needs to be erected, can be made by the count of the number of consumer connections linked with a substation transformer. How to get that count? This count can be easily arrived by the use of counting principles.

In this Chapter, the art of counting is discussed starting with the Fundamental principles of counting, travelling through Permutation and Combinations.

Learning Objectives

On completion of this chapter, the students are expected to know

- the principles of counting and applying it to various situations.
- how to compute the number of ways in arranging a set of distinct objects.
- how to compute the number of arrangements from a set containing identical objects.
- how to compute and applying the strategies to find the number of combinations of a set of different objects.
- the applications of the principle of mathematical induction.

We shall start the chapter with the section on

4.2 Fundamental principles of counting

1. **The Sum Rule** Let us consider two tasks which need to be completed. If the first task can be completed in M different ways and the second in N different ways, and if these cannot be performed simultaneously, then there are $M + N$ ways of doing either task. This is the sum rule of counting.

Example 4.1 Suppose one girl or one boy has to be selected for a competition from a class comprising 17 boys and 29 girls. In how many different ways can this selection be made?

Solution:
The first task of selecting a girl can be done in 29 ways. The second task of selecting a boy can be done in 17 ways. It follows from the sum rule, that there are 17+29 = 46 ways of making this selection.



The sum rule may be extended to more than two tasks. Thus if there are n non-simultaneous tasks $T_1, T_2, T_3, \cdots, T_n$ which can be performed in $m_1, m_2, \cdots, m_n$ ways respectively, then the number of ways of doing one of these tasks is $m_1 + m_2 + \cdots + m_n$.

2. **The Product Rule** Let us suppose that a task comprises of two procedures. If the first procedure can be completed in M different ways and the second procedure can be done in N different ways after the first procedure is done, then the total number of ways of completing the task is $M \times N$

Example 4.2 Consider the 3 cities Chennai, Trichy and Tirunelveli. In order to reach Tirunelveli from Chennai, one has to pass through Trichy. There are 2 roads connecting Chennai with Trichy and there are 3 roads connecting Trichy with Tirunelveli. What are the total number of ways of travelling from Chennai to Tirunelveli?